BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 011 · CPU Performance: Clock Speed, Cache and Cores

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the MAR hold?

Q2. Why is primary storage needed?
A. To display images
B. To connect network devices
C. To hold data in use
D. To print documents

Q3. What happens to RAM and ROM contents when power is removed?

Q4. A computer needs startup instructions when switched on. “ROM normally holds all unsaved user work while applications run.” Correct this. What could happen?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

clock speed — Number of CPU clock cycles per second.

cache — Small, fast memory holding frequently used information.

core — CPU processing unit able to execute instructions.

parallel processing — Running work simultaneously using multiple processing units.

New learning

Clock speed: Clock speed is measured in Hz: clock cycles per second. One cycle does not necessarily equal one instruction.

Cache size: Cache stores frequently needed information closer to the CPU, reducing time spent accessing RAM.

Core count: Multiple cores enable more work in parallel when software supports simultaneous tasks.

Overall comparison: Architecture and task type also affect performance; do not infer an absolute winner from one specification.

Worked example

Compare CPU A (2 GHz, 4 cores, 4 MB cache) with B (3 GHz, 2 cores, 8 MB cache). Which is faster?

1. CPU B has the higher clock speed.

2. CPU A has more cores, which can help with suitably parallel work.

3. CPU B has more cache, potentially reducing RAM access.

4. Explain that workload and architecture determine real performance.

5. Conclude neither is automatically fastest for every task.

We Do · Practise with guidance

Guided 1

CPU C is 2 GHz and CPU D is 3 GHz. Identify which has a higher clock speed.

Worked method: CPU D.

Guided 2

Explain a performance benefit of cache.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The address of the memory location being accessed.

Q2. C — To hold data in use

Q3. RAM loses its contents; ROM keeps its contents because it is non-volatile.

Q4. ROM keeps firmware or startup instructions without power. These let the computer start when switched on.

I Do / We Do pages · Guided practice

Q1. CPU D.

Q2. Frequently needed data can be accessed faster, reducing slower RAM access.

Q3. An application using parallel threads or several programs running concurrently.

You Do pages · Code First

```
clockGHz = 3
seconds = 2
print(clockGHz * seconds)
```

Expected result: 6

You Do pages · Independent questions

Q1. Any two from clock speed, cache size, number of cores [1 each].

Q2. Holds frequently used data/instructions near CPU [1], reducing slower RAM accesses [1].

Q3. More parallel capacity [1]; only when software uses multiple threads [1]; single-threaded code cannot fully use extra cores [1]; other factors such as cache and architecture matter [1].

Exit Ticket · Model answer

False

Try the question again

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Worked method: Frequently needed data can be accessed faster, reducing slower RAM access.

Guided 3

Give a scenario in which more CPU cores help.

Worked method: An application using parallel threads or several programs running concurrently.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the number of billions of clock cycles.

```
clockGHz = 3
seconds = 2
print(clockGHz * seconds)
```

predict

Question 1

State two characteristics affecting CPU performance. [2 marks]

Question 2

Explain how a larger cache may improve performance. [2 marks]

Question 3

Two CPUs both run at 2 GHz but one has twice as many cores. Explain why it may not always finish a task faster. [4 marks]

Exit Ticket · Answer before checking

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

3 GHz guarantees exactly three billion instructions every second.